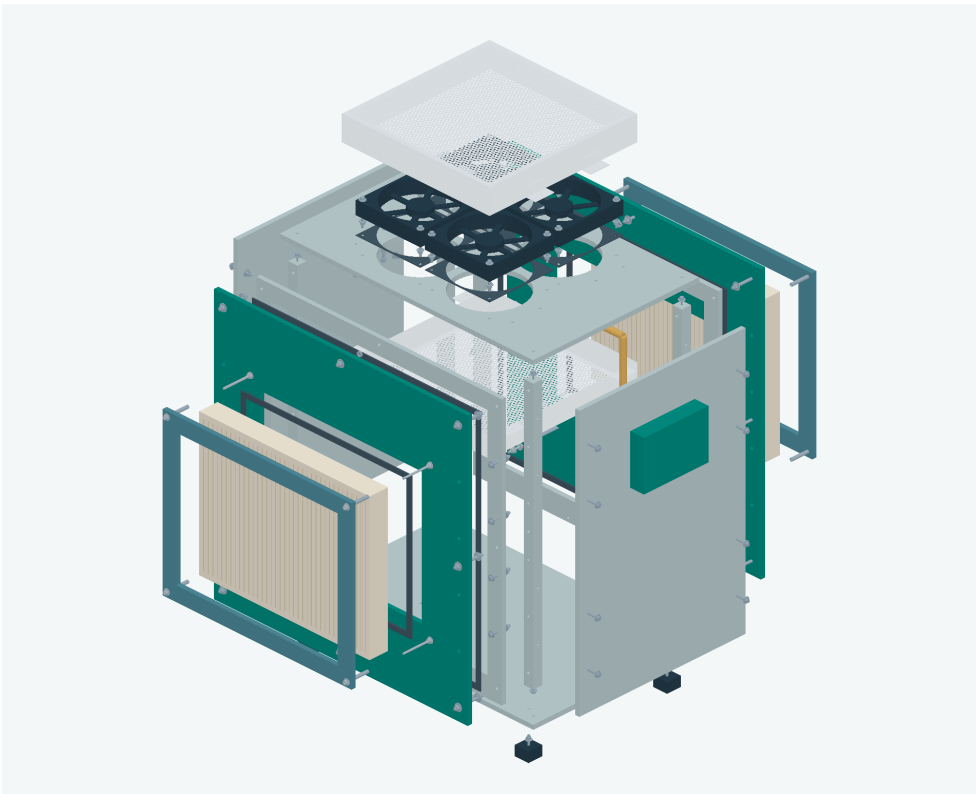


D01 Internal Engineering Handoff

Review revision IH02 | Mechanical R03M | Electrical E00 with supplements | 5 October 2026

Decision requested: review and resolve the open engineering items in REVIEW_AND_RELEASE_REGISTER.csv. This is a coordinated design-review issue, not a fabrication or energization issue. Manufacturer replies are not the only remaining dependency.



Exploded view of the actual current assembly. Omitted threads, weld beads and perforations limit what CAD checks establish. Nominal component fits do not establish strength, safety or sealing.

Evidence class	Meaning
DETECTED	Recorded files or geometry; not physical measurement
OEM REFERENCE	Published component data; not approval of our integration
ASSUMPTION	Provisional input requiring validation
UNKNOWN / OPEN	Do not order, cut or energize the affected interface

Authoritative files and corrections

Use the native R03M assembly as the base. Harness review CAD is a derivative with assumed cable reservations, not a competing machine. The package preserves source-relative paths under engineering/ so scripts can find their inputs.

Use	Controlling source
Mechanical geometry	mechanical_package/D01_R03M_ASSEMBLY.FCStd and STEP;15 DXF files include10 panels plus5 guard blank types
Electrical	electrical_package PDF, BOM and interfaces plus protective selection supplement; no completed rated circuit
Current pressure	CURRENT_PRESSURE_BUDGET.json generated in this handoff from R03M guard area and archived OEM numeric fan points
Current airflow analyzer	engineering/scripts/analysis/d01_validation.py; mandatory --plane-area-m2;23 checks
Latest open decisions	REVIEW_AND_RELEASE_REGISTER.csv and guard/harness/protective supplements

Superseded assumptions: do not use old Filtrete/MERV13 arrangements, large-tower230V fan wiring, earlier long studs or riveted guard details. D01 currently uses two experimental STARKVIND envelopes and four12V P14 Max fans. No certified whole-device filtration class is assigned.

Pressure correction:14.32Pa per filter at an assumed300m3/h total and reducer K1, not the older14.61Pa. This is remaining model headroom, not a measured filter loss or acceptance margin. Old R01 operating-flow predictions do not transfer to STARKVIND.

Appendices: earlier mechanical18-page, electrical7-page and validation5-page PDFs are included as labeled reference snapshots after this review section. The current notes and scripts supersede their outdated analyzer command/test count and any older pressure interpretation. No design release is implied.

Mechanical review disposition

DETECTED:493 objects in the base CAD;119805 checked non-reservation pairs with no positive-volume clash in the saved generator audit. The panel audit reports119804 because it also excludes the guard pair. These are stored geometry results; native CAD hash is verified in this handoff.

Interface	Review action
Filters and seals	370x290x40 filter envelopes; frame bearing and thickness tolerances unverified. Assumed39/40/41mm thickness changes nominal4mm gasket compression from0/25/50%. Do not cut spacers from that assumption.
Cabinet and supports	Panel/cleat grades and joint capacity not selected. Partial assumed mass12.19kg excludes real fans, filters, controls, feet and seals. Do not call it total mass or approve tipping from it.
Guarding	Independent welded guards remain fitted during filter removal.4mm holes/6mm pitch are proposed geometry, not access approval.22/50mm guard-to-body distances are not blade clearance.
Fasteners	Use the R03M schedule and modeled inventory together. One schedule row groups8 adapter studs plus4 ledge screws; do not read its qty8 as the whole hardware count. Grade, preload and wood bearing remain open.
Service	Two200mm filter withdrawal sweeps passed nominal geometry checks. Hands, tools, retainer handling, sag and manufacturing tolerances still require review.

Guard supplement:36 ideal solid-strip cases and28 arithmetic tests explain thickness/span trade-offs only. They do not model the actual2822 holes, plate action, yielding or weld flexibility. A thicker guard or stiffener is not selected. Review loading criteria and unpowered sample assessment.

Electrical review disposition

Reference chain only: OEM NV-PS1 external adapter -> supplied NA-AC10 -> NA-FC1 -> NA-FH1 four-pin input -> four P14 Max fans. SATA stays unused. Instrument logger power remains separate. This is not a terminal-level wiring instruction.

Issue	Required resolution
Power budget	Four fans at0.35A each give1.4A/16.8W nominal.2A/24W path leaves0.6A before controls and startup. A1.5x fan startup scenario alone is2.1A; this is a sensitivity, not a measured surge.
Protective behaviour	PWM zero and hub fuse are not isolation. Hub self-reset and port1-only upstream RPM cannot establish independent four-fan safety monitoring.
Control candidates	Pilz X5/X9P/s4 screens do not select a protective assembly. X9P leaves0.2W before auxiliaries; s4 needs24V; X5 monitored sequence not established.
Enclosure and harness	140x100x40mm is reserved space, not a selected box. Include walls, terminals, bend radii, cooling, glands and access after circuit selection.
Review output	Issue rated schematic, interface/terminal and cable schedule, protection coordination, enclosure layout and commissioning procedure. Retain or explicitly revise restart requirements through hazard review.

Keep OEM mains adapter intact. No bare-mains design is released. Internal electrical engineer determines applicable Indian requirements, supply/site suitability and required inspection. An external adapter rating does not certify this assembly. No arbitrary fuse size, wire gauge or terminal mapping has been filled in.

The14 named control cases and1024 one-step states are software logic checks only. They do not test hardware timing, welded contacts, sensing coverage, DC interruption, thermal behaviour or safety integrity.

Pressure and measurement basis

Four fans operate in parallel: flow adds, pressure does not. Two filter branches operate in parallel: each ideally sees half total flow; their pressure drops are not added. Unequal branch flows remain possible in real construction.

Total m3/h	Each filter m3/h	Fan Pa	Other losses Pa	Filter allowance Pa
200	100.0	32.60	5.92	26.68
250	125.0	30.09	9.24	20.85
300	150.0	27.63	13.31	14.32
350	175.0	25.27	18.12	7.15
400	200.0	22.83	23.66	-0.83

ASSUMPTIONS:2800rpm OEM curve; density1.2kg/m3; equal fan/filter sharing; guard area0.0354623m2 each; guard K1.5 each, plenum K1.5, installation K1 and reducer K1. Full JSON also gives reducer K0.5/2 sensitivity. Coefficients, static/installed compatibility and any double counting remain unverified. Negative headroom means that scenario has no allowance for a positive filter loss.

No actual STARKVIND clean/loaded curve has been established. Do not promise300m3/h, CADR, HEPA-level removal or room coverage. Fit and pressure evidence are separate gates.

Current airflow tool requires accepted physical rows, unique cells, one run/instrument/method/calibration, ISO timestamps with timezone and independent full-plane area. Equal cell-area sum does not prove correct spatial coverage. Reported error is velocity-only; it excludes area, swirl, fixture effects and time variability.

Run from engineering/: **python scripts/analysis/d01_validation.py**. With real reviewed data: add **--airflow-csv FILE --plane-area-m2 AREA --output NEW.json**. AREA is the measurement plane, not automatically guard open area. Never alter raw observations to obtain a passing result.

Review ownership and release gates

IDs	Owner	Closure package
M01-M03	Mechanical engineer	Guard/structure/seal calculations, selected materials/joints, corrected drawings and inspection plan
M04	Mechanical + electrical	OEM reply interpretation and protected harness integration
E01-E03	Electrical + mechanical	Hazard allocation, rated circuit/enclosure and commissioning evidence
P01 / T01-T02	Test lead + project lead	Filter/duty evidence, agreed room/targets, calibrated methods and physical results
C01	Project lead + engineers	Real quotes and released revision-specific BOM/build authorization
O01	Future outdoor team	Separate public/weather/wind/stability design and field evidence

Complete the CSV register with reviewer name, disposition, date and linked evidence. A row is not closed by ticking a box: the accepted calculation, drawing or physical result must exist. No reviewers or approvals have been invented.

What can happen now: internal drawing/calculation review, applicability/hazard decisions, manufacturability discussions and scoped quotations if separately authorized. These do not require the manufacturer cable reply.

Waiting: cable/connector evidence from ARCTIC and Noctua; Noctua ticket220175. Filter/frame/curve data, site selection and actual physical evidence are separate unresolved inputs, not covered by those enquiries.

Release boundary: only a signed revised package with closed safety-critical holds can authorize affected fabrication/assembly. Digital work, supplier delivery, workshop fabrication, inspection and physical testing are distinct critical-path activities. No completion date is promised.

Assembly and verification route

Sequence below applies only after component-specific release. Use the mechanical appendix for drawing details; unresolved quantities/ratings must not be guessed on the workshop floor.

Stage	Action and hold point
1 Receive and inspect	Record exact fan/filter/adaptor revisions, dimensions and condition. Reject unapproved substitutions; check against released BOM.
2 Unpowered dry fit	Inspect cut edges, panel joints, feet, fan mount and guard fit. Keep electronics disconnected. Resolve tolerance clashes before sealing.
3 Seals and filters	Fit reviewed ledges, penetration seals, gaskets, stops and retainers in the documented order. Verify compression/flatness and supported service access.
4 Guards and wiring	Fit independently retained guards; install released harness restraints and enclosure. Verify cable isolation from moving parts and guarded openings.
5 Commission	Qualified electrical/mechanical inspection and approved isolation/stop/restart/thermal checks. Record sign-off before performance operation.
6 Measure	Record airflow and pressure at reviewed settings; power and noise; controlled room PM decay/background only under approved method. No improvised smoke, powder or gas challenge.
7 Decide	Compare to agreed targets, keep failed results, investigate leakage/low flow/noise. Update design and repeat relevant verification before expansion.

Blank CSVs are included. Instrument ranges, calibration and fixture loading must match the approved test method. Consumer PM readings do not certify filter class. Do not infer outdoor radius from an indoor decay test.

Parts funding and evidence handover

COMBINED_PARTS_REGISTER.csv separates exact OEM references, custom geometry and unresolved hardware/services. CAD_PART_INVENTORY.csv is a modeled-object list, not an order list. FASTENER_SCHEDULE.csv groups hardware; do not sum both as additional purchases.

Use staged funding: review and engineering closure; then released components/fabrication; then commissioning and calibrated testing. Request actual quotes and lead times against the approved revision. No INR total is asserted without quotations.

The ZIP includes selected current native/STEP/exploded files, 15 DXFs, current engineering evidence and standalone calculation inputs. No old CAD branches, downloaded manufacturer manuals, email address or conversation records are included. OEM facts must still be checked against delivered parts.

Sources are recorded in electrical_package/sources.json, harness_detail/README.md and GUARD_ACCESS_DECISION.md. Principal official sources include ARCTIC P14 Max support, Noctua NV-PS1/NA-FC1/NA-FH1 manuals and IKEA India STARKVIND104.633.30. Archived references are evidence, not current stock or price guarantees.

Current filter comparison: scripts/analysis/d01_current_filter_screen.py uses the current R03M budget. Supply per-filter m3/h and Pa with exact part/revision, source, CLEAN or LOADED condition and its conditioning reference. It refuses extrapolation and preserves input hashes. Positive model margin is conditional, not approval. Do not use the earlier R01 offer checker for this design. See TOOL_USAGE.md.

The geometric assembly is ours; proprietary purchased parts retain their manufacturers' rights. This package is for internal review and does not assert patentability or a license to redistribute OEM materials.

Handback requested from the team: one annotated review register and one coordinated revision, not parallel incompatible assemblies. Keep changes traceable to R03M and record any accepted change to safety requirements. Do not mark all remaining work complete while these decisions remain open.

Annexes follow: mechanical R03M; electrical E00; validation V00. All are reference snapshots and remain review-only. Corrections in this review section take precedence. The complete machine has not been built or tested.